3.2.1
Find solution using Simplex method (BigM method)
MIN Z = 5x1 + 3x2
subject to
2x1 + 4x2 >= 16
4x1 + 3x2 >= 24
and x1,x2 >= 0

Solution:
The problem is converted to canonical form by adding slack, surplus and artificial variables as appropriate
1. As the constraint-1 is of type '≥' we should subtract surplus variable S_1 and add artificial variable A_1
2. As the constraint-2 is of type '≥' we should subtract surplus variable S_2 and add artificial variable A_2

After introducing surplus, artificial variables.

$\text{Min } Z = 5x_1 + 3x_2 + 0S_1 + 0S_2 + MA_1 + MA_2$

subject to

$2x_1 + 4x_2 - S_1 + A_1 = 16$

$4x_1 + 3x_2 - S_2 + A_2 = 24$

and $x_1, x_2, S_1, S_2, A_1, A_2 \geq 0$

Iteration-1

Cj	***Basic Variables***	***Quantity***	5	3	0	0	*M*	*M*	**Min Ratio** ***Quantity/ pivot column***
			x1	***x2***	***S1***	***S2***	***A1***	***A2***	
M	A1	*16*	2	**(4)**	-1	0	1	0	16/4=4→
M	A2	*24*	4	3	0	-1	0	1	24/3=8
	Zj	**40*M***	**6*M***	**7*M***	***-M***	***-M***	***M***	***M***	
	Zj-Cj		6*M*-5	7*M*-3↑	*-M*	*-M*	0	0	

Positive maximum $Z_j - C_j$ is $7M-3$ and its column index is 2. So, the entering variable is x_2.
Minimum ratio is 4 and its row index is 1. So, the leaving basis variable is A_1.

∴ The pivot element is 4.
Entering $=x_2$, Departing $=A_1$, Key Element =4
$R_1(\text{new}) = R_1(\text{old}) \div 4$

$R_1(\text{old}) =$	16	2	4	-1	0	0
$R_1(\text{new}) = R_1(\text{old}) \div 4$	4	0.5	1	-0.25	0	0

$R_2(\text{new})=R_2(\text{old}) - 3R_1(\text{new})$

$R_2(\text{old})$ =	24	4	3	0	-1	1
$R_1(\text{new})$ =	4	0.5	1	-0.25	0	0
$3\times R_1(\text{new})$ =	12	1.5	3	-0.75	0	0
$R_2(\text{new})=R_2(\text{old}) - 3R_1(\text{new})$	12	2.5	0	0.75	-1	1

Iteration-2

Cj	***Basic Variables***	***Quantity***	5	3	0	0	*M*	**Min Ratio** ***Quantity/ pivot column***
			x1	***x2***	***S1***	***S2***	***A2***	
3	*x2*	4	0.5	1	-0.25	0	0	4/0.5=8
M	A2	12	**(2.5)**	0	0.75	-1	1	12/2.5=4.8→
	Zj	**12M+12**	**2.5*M*+1.5**	**3**	**0.75*M*-0.75**	***-M***	***M***	
	Zj-Cj		2.5*M*-3.5↑	0	0.75*M*-0.75	*-M*	0	

Positive maximum Z_j-C_j is 2.5*M*-3.5 and its column index is 1. So, the entering variable is x_1.
Minimum ratio is 4.8 and its row index is 2. So, the leaving basis variable is A_2.

∴ The pivot element is 2.5.
Entering =x_1, Departing =A_2, Key Element =2.5
$R_2(\text{new})=R_2(\text{old}) \div 2.5$

$R_2(\text{old})$ =	12	2.5	0	0.75	-1
$R_2(\text{new})=R_2(\text{old}) \div 2.5$	4.8	1	0	0.3	-0.4

$R_1(\text{new})=R_1(\text{old}) - 0.5R_2(\text{new})$

$R_1(\text{old})$ =	4	0.5	1	-0.25	0
$R_2(\text{new})$ =	4.8	1	0	0.3	-0.4
$0.5\times R_2(\text{new})$ =	2.4	0.5	0	0.15	-0.2
$R_1(\text{new})=R_1(\text{old}) - 0.5R_2(\text{new})$	1.6	0	1	-0.4	0.2

Iteration-3

Cj	***Basic Variables***	***Quantity***	5	3	0	0	**Min Ratio** ***Quantity/ pivot column***
			x1	***x2***	***S1***	***S2***	
3	*x2*	1.6	0	1	-0.4	0.2	---
5	***x1***	4.8	1	0	**(0.3)**	-0.4	4.8/0.3=16→
	Zj	**28.8**	**5**	**3**	**0.3**	**-1.4**	
	Zj-Cj		0	0	0.3↑	-1.4	

Positive maximum Z_j-C_j is 0.3 and its column index is 3. So, the entering variable is S_1.
Minimum ratio is 16 and its row index is 2. So, the leaving basis variable is x_1.

∴ The pivot element is 0.3.

Entering $=S_1$, Departing $=x_1$, Key Element $=0.3$

$R_2(new)=R_2(old) \div 0.3$

$R_2(old)$ =	4.8	1	0	0.3	-0.4
$R_2(new)=R_2(old) \div 0.3$	16	3.3333	0	1	-1.3333

$R_1(new)=R_1(old) + 0.4R_2(new)$

$R_1(old)$ =	1.6	0	1	-0.4	0.2
$R_2(new)$ =	16	3.3333	0	1	-1.3333
$0.4 \times R_2(new)$ =	6.4	1.3333	0	0.4	-0.5333
$R_1(new)=R_1(old) + 0.4R_2(new)$	8	1.3333	1	0	-0.3333

Iteration-4

Cj	***Basic Variables***	***Quantity***	5	3	0	0	**Min Ratio** ***Quantity/ pivot column***
			x1	***x2***	***S1***	***S2***	
3	x_2	8	1.3333	1	0	-0.3333	---
0	S_1	16	3.3333	0	1	-1.3333	4.8/0.3=16→
	Zj	**24**	**4**	**3**	**0**	**-1**	
	Z_j-C_j		-1	0	0	-1	

Since all Z_j-$C_j \leq 0$

Hence, optimal solution is arrived with value of variables as :

$x_1=0, x_2=8$

Min $Z=24$